

Safaet Jaman Arman

AI/ML Engineer | LLM & RAG Systems | NLP Researcher

Chattogram, Bangladesh | esafaetjaman@gmail.com | +8801306726612 | github.com/safaet | linkedin.com/in/safaet-jaman | safaet.github.io/Personal_portfolio

SUMMARY

AI/ML Engineer with production experience shipping RAG systems for multilingual (Bengali/romanized-Bengali/English) conversational products. Designed a two-stage retrieval pipeline (bge-m3 bi-encoder + bge-reranker-v2-m3 cross-encoder) with a data-calibrated refusal gate, RAGAS-based evaluation harness, and LLM tracing/observability, while leading a 4-person delivery team. Co-author of a peer-reviewed antimicrobial resistance (AMR) prediction paper accepted at ECCT 2026 (with icddr,b), introducing a novel Resistome Burden Index (RBI) feature. Background spans full-stack ML engineering (FastAPI, PostgreSQL, Docker) and applied ML research with rigorous statistical validation.

TECHNICAL SKILLS

Languages: Python, SQL

ML/DL Frameworks: PyTorch, HuggingFace Transformers, PEFT/LoRA, scikit-learn, XGBoost, LightGBM, CatBoost, AMP (mixed precision)

LLM & NLP Tooling: LangChain, LangGraph, FAISS, ChromaDB, RAGAS, BanglaBERT, XLM-RoBERTa, mBERT, Prompt Engineering, Intent Routing, Tool-Use Agents

MLOps / Infra: Docker, systemd, GitHub Actions, Weights & Biases, MLflow, HuggingFace Hub, vLLM, AWS

Backend / Data: FastAPI, Django, PostgreSQL, SQLAlchemy, Pydantic v2, Async Request Handling

Research & Evaluation: McNemar's Test, Bootstrap CI, Wilcoxon Signed-Rank Test, LaTeX

EXPERIENCE

Technical Lead, AI/ML — Neuralis AI (Bengali RAG Chatbot, Client Project)

Part-time | Aug 2026 – Present

- Led a 4-person cross-functional team (frontend, 2× data-collection, deployment) end-to-end while owning the ML/backend implementation directly
- Architected a two-stage retrieval pipeline (bge-m3 bi-encoder + bge-reranker-v2-m3 cross-encoder) with a data-calibrated refusal gate to suppress hallucinated answers on out-of-scope queries
- Built the evaluation/observability stack: RAGAS-based faithfulness/relevancy scoring, chunking A/B test harnesses, and self-hosted LLM tracing for per-stage token cost, latency, and quality
- Owned the GitHub repository and release process, enabling independent deployment from main with zero handoff friction

AI Engineer — Neuralis AI (Multilingual Conversational AI Product)

Part-time (~25 hrs/wk) | Prior to Aug 2026

- Built a full retrieval pipeline — LLM-based query rewriting (romanized/English to target script, with caching + fallback), bge-m3 multilingual embeddings (1024-dim) with custom chunking, and cross-encoder reranking
- Implemented score-gated grounded generation with context-only prompting and inline citations, refusing unsupported queries instead of hallucinating
- Designed a regex-first intent classifier with LLM fallback, routing a single endpoint to grounded QA, map-reduce summarization, persona chat, and recommendation services
- Built a normalized PostgreSQL schema with incremental ingestion, per-record embedding-state tracking, and stale-chunk cleanup for safe re-indexing
- Served via async FastAPI (blocking calls offloaded to threadpool), with typed schemas and per-request latency/retrieval-score telemetry
- Containerized deployment (Docker, systemd); built offline eval tooling for retrieval-threshold tuning and embedding regression tests

Junior AI/ML Engineer — Code Owls (Generative AI / LLM Automation)

Full-time | Jul 2024 – May 2025

- Developed and deployed GenAI workflows using LangChain and LangSmith for dynamic, NLP-driven LLM response pipelines
- Built and maintained AI automation solutions (Python, FastAPI, Git, Linux) across data preprocessing, model integration, and API deployment
- Applied structured A/B evaluation and feedback loops in an Agile cross-functional team to improve model accuracy and UX

RESEARCH

Research Collaborator — icddr,b

- Developed an SVM-XGBoost ensemble for AMR phenotype prediction on clinical genomic data; introduced the Resistome Burden Index (RBI) as an interpretable composite feature
- Ran full statistical significance testing (McNemar's, Bootstrap CI, Wilcoxon signed-rank) for model comparison

Publication: AMR Prediction via SVM-XGBoost Ensemble with Resistome Burden Index — ECCT 2026, accepted (camera-ready submitted), in collaboration with icddr,b. Repository: github.com/safaet/AMR-Prediction

SELECTED PROJECTS

ResistomeX — AMR Prediction Web App

Python, Streamlit, scikit-learn, XGBoost

- Productized the R-Blend ensemble (Decision Tree + LogReg + XGBoost, soft voting, RBI feature) into a live tool; avg. F1 of 0.9421 across 12 antibiotic-pathogen datasets
- Live: resistomex.streamlit.app | GitHub: github.com/safaet/ResistomeX

EDUCATION

BSc, Computer Science & Engineering — International Islamic University Chittagong (IIUC)

2019 – 2026 (Expected)

Relevant coursework: DSA, Machine Learning, Deep Learning, NLP, Computer Vision, Statistics & Probability, Linear Algebra

ACHIEVEMENTS

- Co-authored peer-reviewed paper accepted at ECCT 2026 — first ML-based AMR prediction study on the icddr,b dataset
- 4th Place, IIUC Intra-University Programming Contest, Autumn 2022